

(Please write your Enrollment Number)

Enrollment No. _____

REMID-TERM EXAMINATION
(Course Name: B.Tech) (Semester: 1st)
(November 2024) OFF LINE mode

Subject Code: BCS 101	Subject: Programming with C
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
a)	Define the following terms in C: Variables, Data Types, and Operators. Provide one example for each.		CO1
b)	Analyze the following C code snippet. Explain the purpose of each part of the code and determine the final output: <pre>int x = 8, y = 3, z; z = x++ + --y * 2; printf("%d", z);</pre>		CO1
c)	Examine the C code below and explain the output. Discuss the use of nested loops and how they influence the output pattern: <pre>for (int i = 1; i <= 3; i++) { for (int j = 1; j <= i; j++) { printf("*"); } printf("\n"); }</pre>		CO2
d)	Given the following code, analyze the output and describe the role of character manipulation and string used: <pre>char str[] = "Hello"; for (int i = 0; i < strlen(str); i++) { printf("%c ", str[i] + 1); }</pre>		CO2
UNIT I			
Q2	Attempt any two parts	(5*2=10)	CO Mapping
a)	Discuss the purpose and use of the shift operators in C programming. Provide examples of both left shift and right shift operations, explaining the effect on integer values.		CO1
b)	Predict the output of the following code and analyze the effect of operator precedence and associativity on the expression's evaluation: <pre>int a = 10, b = 20, c = 30, result; result = a + b * c / a - b + c % a; printf("%d", result);</pre>		CO1
c)	Explain the concept of type casting in C and differentiate between implicit and explicit type casting. Provide examples showing how each type of casting affects the results of expressions.		CO1
UNIT II			
Q3	Attempt any two parts	(5*2=10)	CO Mapping
a)	Compare the efficiency of for, while, and do-while loops in C programming. Discuss when each type of loop is best suited based on different use cases and performance considerations.		CO2

b)	Predict the output of the following code, which manipulates a 2D array. Explain each step and describe how elements are accessed and modified in a 2D array: <pre>int arr[2][3] = {{1, 2, 3}, {4, 5, 6}}; for (int i = 0; i < 2; i++) { for (int j = 0; j < 3; j++) { arr[i][j] = arr[i][j] + i + j; printf("%d ", arr[i][j]); } printf("\n"); }</pre>		CO2
c)	Write a C program that takes a positive integer from the user and determines if it is a prime number. Provide comments explaining each part of your code.		CO2